

- Machine learning has changed the way data interpretation and extraction works by involving several automatic sets of generic methods, which have replaced statistical techniques.
- It provides insights that help to create applications that are more intelligent and data-driven, and hence improves their operation and business processes, leading to easier decision making.
- Machine learning software systems improve their performance more and more, as people use them. This occurs because the algorithms used in them learn from the large set of data generated on the basis of the users' behaviour.
- It helps in inventing new ways to solve some sudden and abrupt challenges in the system, on the basis of the experience gained by the machine while analysing the large data sets and behaviour of the system.
- The increasing use of machine learning in industries acts as a catalyst to make data science increasingly relevant.

Some of the machine learning techniques used in data science

- Decision tree learning:** This is a machine learning technique that uses a decision tree as the predictive model, which further maps observations about an item to the conclusions about the target value of the item.
- Association rule learning:** This is a method used for discovering several interesting relations between the variables in large databases.
- Artificial neural networks:** Such learning techniques are also called neural networks. These are learning algorithms that are inspired by the structure and functional aspects of biological neural networks. Different computations are actually structured in terms of interconnected groups of artificially designed neurons, which help to process information using the connectionist approach to computation. All the modern neural networks are basically non-linear statistical tools used for data modelling. They are usually used to model several complex relationships between the inputs-outputs and to find patterns in the data.
- Inductive logic programming (ILP):** This approach uses logical programming as a representation for several input examples, the background knowledge and the hypotheses. If we are given an encoding of any known background knowledge with a set of examples, which represent a logical database of facts, then an ILP system will easily derive a hypothesised logic program which entails all the positive and no negative examples. This type of programming considers any type of programming language for representing the hypotheses, such as functional programs.
- Clustering:** Cluster analysis is a technique used for the assignment of a set of different observations into various subsets (also called clusters) so that the observations



Figure 5: Machine learning and data science (Image credit: googleimages.com)



Figure 6: The machine learning process (Image credit: googleimages.com)

present within the same cluster are similar to some pre-designated criteria, whereas the observations drawn from all the different clusters are dissimilar. All the different clustering techniques have different assumptions on the structure of data, which is often defined by some similarity metric and is evaluated, for example, by internal compactness and the separation between different clusters. It is basically an unsupervised learning method and one of the common techniques used for statistical data analysis.

- Bayesian networks:** A Bayesian network is a probabilistic graphical model which represents a set of random variables and all their conditional independencies using a directed acyclic graph. For instance, a Bayesian network can represent the probabilistic relationships between different diseases and their symptoms. If we are given the symptoms, then the network can easily compute the probabilities of the presence of various diseases. Very efficient algorithms are used to perform the inference and learning.
- Reinforcement learning:** This is a technique related to how an agent ought to take different actions in an environment in order to maximise some notion of the long-term reward. This type of algorithm attempts to find a policy that maps different states of the world to the different actions the agent ought to take in those states. This type of learning differs from the supervised learning problem, for which correct input/output pairs are not presented, nor are the sub-optimal actions explicitly corrected. 

References

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